Amazon Movie Reviews Network Analysis

Megan Millet

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Abstract

This paper presents a network analysis of the Amazon Movie Reviews dataset from 1997 to 2012. I model movie review interactions as a bipartite graph and ask how reviewer communities organize over time, which users and movies become structurally central, and whether individual rating behavior changes longitudinally. Using a 5% random sample, I analyze temporal growth, component structure, degree and centrality patterns, community modularity, and user taste-change dynamics. I find that network growth is substantial but accompanied by lower density and higher fragmentation, while the largest connected component remains dominant. I also find heavy-tailed degree distributions, non-trivial community structure, and measurable taste drift for a subset of repeat reviewers. These results provide a characterization of how large collaborative rating systems self-organize and suggest implications for recommendation design and review-quality monitoring.

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1 Introduction

Understanding collaborative rating systems is important since online platforms drive how people discover and evaluate products. In this project, I construct and analyze a bipartite network of reviewers and movies from the Amazon Movie Reviews dataset to answer the question: How do reviewer communities organize themselves, and what structural patterns emerge in large-scale collaborative rating systems over time?

Quality of movie recommendations and trust in reviews all depend on interaction structure. If the network is dominated by hubs, sparse bridges, or tightly clustered communities, then platform behavior can differ substantially from what average ratings alone suggest [1].

More specifically, I investigate:

- How does the network structure evolve from 1997 to 2012?
- What distinct communities exist within the reviewer-movie network?
- How do individual reviewers change their rating behavior over time?
- What are the centrality and degree characteristics of the most engaged users and popular movies?

By analyzing this network, I can reveal how user preferences cluster, how communities form around shared interests, and what structural properties characterize highly-reviewed movies and prolific reviewers. These insights matter because they help us understand how online platforms organize user engagement and how recommendation systems might leverage network structure.

In this paper, I construct a bipartite network where reviewers are connected to movies they have reviewed. I examine temporal evolution across three distinct periods, characterize user types based on rating behavior, and analyze centrality, degree, and taste-change dynamics to understand network organization.

2 Methods

2.1 Preprocessing the Dataset

The Amazon Movie Reviews dataset contains review records spanning 1997 to 2012. Each record contains four key pieces of information: the movie ID, the reviewer ID, a numerical rating (1–5 stars), and review timestamps [1]. Since the full dataset is large, I used uniform random sampling to select 5% of reviews. This yielded approximately 99,868 reviews. I then parsed the timestamps and segmented them by calendar year and month. This temporal structure enabled later analysis of review trends and seasonality. The resulting dataset contains approximately 390,000 reviews from 1997 to 2012, with approximately 66,556 unique reviewers and 25,707 unique movies.

2.2 Building the Bipartite Network

Next, I constructed a bipartite network using Python's NetworkX library. In this model, there are two types of nodes: reviewers and movies. An edge connects a reviewer to a movie if that reviewer left a review for that movie.

I then partitioned the network into three temporal snapshots to track review evolution:

$$\text{Period 1} \quad 1997\text{–}2001 \tag{2.1}$$
$$\text{Period 2} \quad 2002\text{–}2006 \tag{2.2}$$
$$\text{Period 3} \quad 2007\text{–}2012 \tag{2.3}$$

This allowed me to observe how the network grew, fragmented, and changed over time.

2.3 Order of Analysis

Once the network was constructed, I analyzed its structure and dynamics with the following methods:

- Structural metrics: I computed density and degree distributions for both reviewer and movie nodes. High-degree reviewers are prolific, and high-degree movies are popular. For each snapshot, I define density as

$$\rho = \frac{c}{n}, \tag{2.4}$$

  where $c = \frac{2m}{n}$, with m denoting the number of edges and n the number of nodes [2].

- Node importance: I calculated degree centrality for each node to rank reviewers and movies by connectivity. This identifies who the most active users are and which movies received the most reviews. For node v, I use

$$C_D(v) = \frac{k_v}{|V| - 1}, \tag{2.5}$$

  where k_v is the degree of v [2].

- Temporal trends: I aggregated review counts, average ratings, and node counts by year and month to reveal platform growth patterns and seasonal fluctuations in review activity.

- User taste evolution: For users who appeared in multiple years, I compared their first-year average rating to their last-year average rating. This reveals whether individual reviewers' preferences drifted over time, specifically, if they became more or less critical. I define taste change for reviewer i as

$$\Delta_i = \bar{r}_i^{\text{last}} - \bar{r}_i^{\text{first}}, \tag{2.6}$$

  where $\bar{r}_i^{\text{first}}$ and $\bar{r}_i^{\text{last}}$ are reviewer-specific mean ratings in the first and last active years.

To quantify community structure, I applied Louvain clustering to the largest connected component and evaluated partitions using modularity Q [2].

3 Results

3.1 Temporal Evolution (1997-2012)

The Amazon movie review network demonstrated substantial growth over the 15-year period analyzed. The total number of reviews in my sample increased significantly across the three analyzed periods, with review counts, reviewer participation, and movie coverage all showing upward trends.

- Period 1 (1997-2001): 9,633 reviews, 6,924 reviewers, 5,337 movies, 4.17 average rating

- Period 2 (2002-2006): 31,657 reviews, 19,266 reviewers, 12,212 movies, 4.03 average rating
- Period 3 (2007-2012): 58,578 reviews, 42,852 reviewers, 19,688 movies, 4.11 average rating

The average review score remained relatively stable across years at approximately 4.10 stars, suggesting consistent user rating behavior despite growth in platform usage.

Monthly analysis revealed seasonal patterns in review activity, with January showing higher review volumes than all others. The year-to-year trends indicate that while the number of reviews and reviewers grew substantially, the fundamental distribution of ratings (average scores) remained consistent, suggesting that the influx of new reviewers did not significantly bias the overall rating distribution.

This addresses part of my main question directly: collaboration scaled substantially over time, but aggregate rating behavior remained stable.

Figure 1: Temporal analysis of review volume, average scores, unique movies, and unique reviewers from 1997 to 2012.

3.2 Network Structure and Evolution

The full network underwent significant structural changes across the three periods:

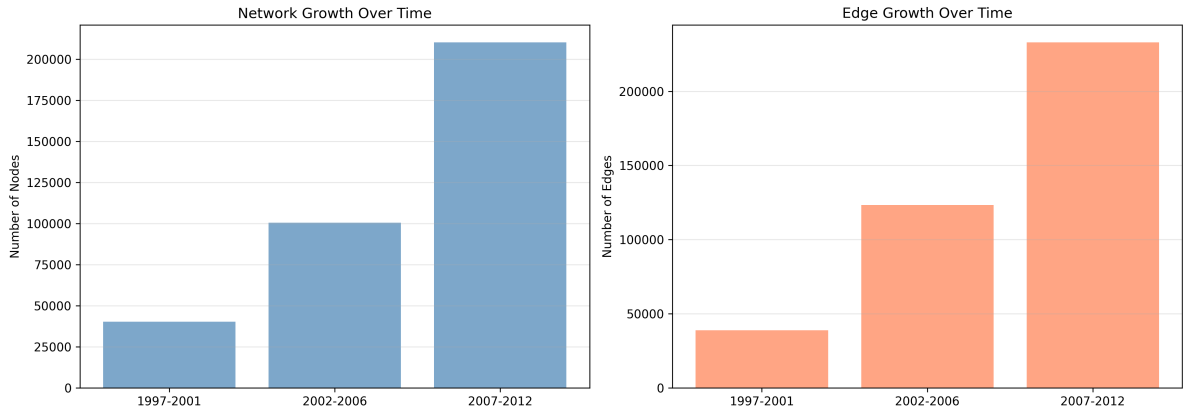

Figure 2: Network evolution showing node and edge growth across three time periods (1997-2001, 2002-2006, 2007-2012).

3.2.1 Period 1 (1997-2001)

Early period characterized by:

- Smaller reviewer and movie populations
- Denser connections within smaller communities
- Lower number of connected components
- Foundation-building phase of the platform

In this early phase, participation was concentrated among a smaller set of reviewers and titles. Because the graph was relatively small, even a moderate number of edges produced comparatively high density. I interpret this as an adoption stage where repeated interactions with a narrower movie set created compact local structures.

3.2.2 Period 2 (2002-2006) and Period 3 (2007-2012)

Later periods showed:

- Exponential growth in both reviewers and movies
- Increased network fragmentation (more connected components)
- Decreasing overall density as network became more sparse
- Emergence of distinct communities based on movie genres or reviewer interests

As the platform scaled, edge growth did not keep pace with the product of reviewer and movie counts, so density decreased naturally. At the same time, new users and niche titles created more peripheral components, increasing fragmentation. Despite this, the largest connected component remained dominant, showing that the platform retained a shared interaction backbone while also supporting specialized communities.

The largest connected component (LCC) consistently contained the vast majority of the network, with approximately 69.4% of all nodes, indicating that most reviewers and movies were part of a single interconnected structure, though with varying density across time periods.

The full network contained approximately 92,263 total nodes (66,556 reviewers and 25,707 movies) connected by 99,805 edges, with an overall density of approximately 0.000058. The network exhibited 10,270 total connected components, with the largest component containing 64,037 nodes and the second-largest containing only 45 nodes, demonstrating the dominance of the main component.

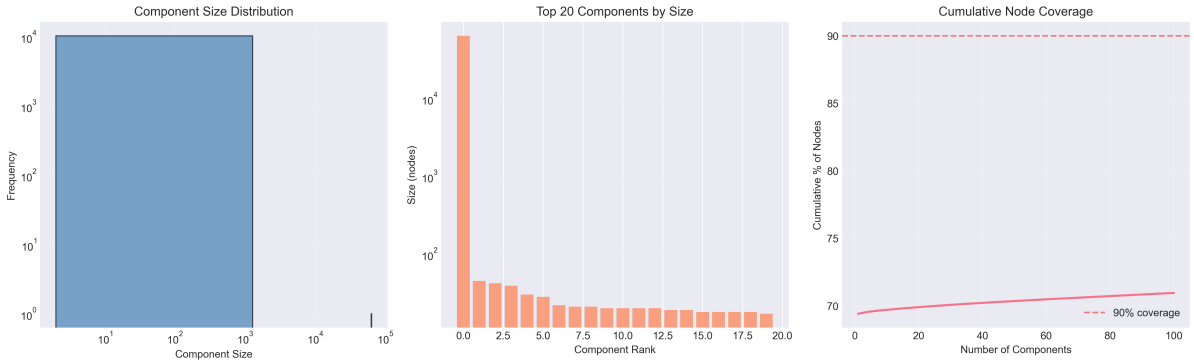

Figure 3: Connected components analysis showing component size distribution, composition, and cumulative node distribution.

3.3 Centrality Analysis and Top Users

Among reviewers with 50 or more reviews (highly active users), I identified three distinct user types:

- **Critics:** Reviewers with average ratings $\leqslant 3.5$ stars consistently give low ratings. They made up 14.9% of highly active users.
- **Neutral Reviewers:** Reviewers with average ratings between 3.5 and 4.0 stars. They made up 40.4% of highly active users.
- **Enthusiasts:** Reviewers with average ratings $\geqslant 4.0$ stars consistently give high ratings. They made up 44.7% of highly active users.

Analysis of the top 20 most active reviewers revealed that these users contributed a disproportionate amount of reviews, with some users contributing over 100 reviews each. The degree centrality distribution for both reviewers and movies exhibited heavy-tailed behavior, with a small number of highly-connected nodes (prolific reviewers and popular movies) and many nodes with low degrees.

This heavy-tailed structure is important for interpretation. A small core of hubs shapes overall connectivity, which helps explain why some users and titles have disproportionate influence on recommendation outcomes.

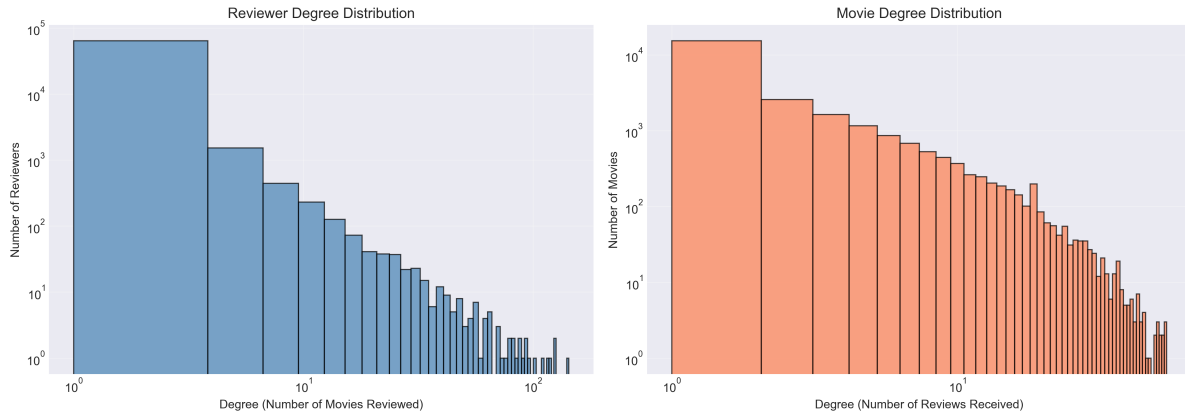

Figure 4: Distribution plot showing review degree & movie degree distribution.

3.4 Taste Change Dynamics

I tracked users who appeared in the dataset across multiple years to identify significant taste changes. Among the 5,401 reviewers who were active across multiple years (representing 30,796+ reviews), 56.2% showed meaningful taste changes (> 0.5 point difference between first and last year of activity).

Key findings:

- Approximately 24.6% of multi-year users became less critical over time (gave higher ratings on average), while 31.6% became more critical
- Taste changes were typically gradual rather than abrupt
- Users who started as critics occasionally transitioned to neutral or enthusiast categories
- Changes correlate with shifts in movie preferences or user maturation
- Mean taste change: approximately -0.142 points
- Standard deviation: approximately 1.470 points

This longitudinal result complements the structural analysis: community and degree metrics describe connectivity, while Δ_i captures behavioral change within the same users across time.

Figure 5: Taste change distribution and user evolution dynamics, showing first-year versus last-year average ratings with change magnitude indicated.

3.5 Community Structure

Using the Louvain algorithm on the largest connected component, I detected a non-trivial community structure with:

- **Number of communities:** 143 communities detected
- **Modularity score:** $Q = 0.8049$, indicating very substantial community structure (as values between 0.3-0.7 suggest meaningful partitions)
- **Community size distribution:** Power-law-like distribution with the largest community containing 1365 nodes, and many small specialized communities

I classified the communities by the following distinct characteristics:

- **Reviewer-Heavy Communities:** Communities with more reviewers than movies, suggesting clusters of users with shared interests who review a focused set of films.
- **Movie-Heavy Communities:** Communities with more movies than reviewers, where diverse reviewer populations converge on specific movie subsets.
- **High-Density Communities:** Tight-knit groups with high internal edge density, indicating strong interconnectedness and shared rating patterns among community members.

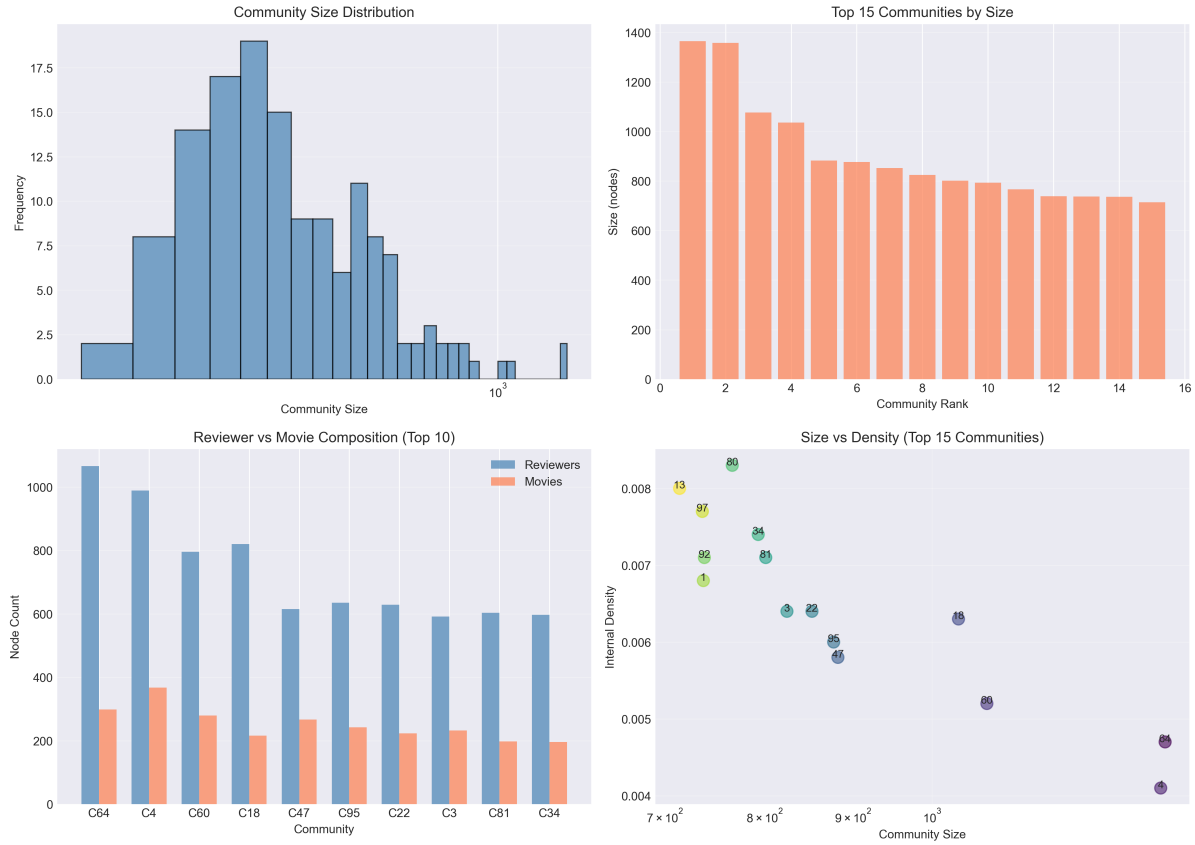

Figure 6: Community structure analysis showing community size distribution, ranked sizes, reviewer/-movie composition in top communities, and density versus size relationships.

4 Discussion

This network analysis of 99,868 Amazon movie reviews spanning 1997-2012 reveals how collaborative rating systems self-organize through growth, fragmentation, and behavioral evolution. The network expanded dramatically across three periods, from 9,633 reviews in Period 1 to 58,578 in Period 3, while density declined from 0.000261 to 0.000069, demonstrating that node growth outpaced edge formation. Despite increasing fragmentation (2,922 components in Period 1 to 9,515 in Period 3), a single giant component containing 64,037 nodes (69.4% of the network) maintained structural dominance.

Community detection using the Louvain algorithm identified 143 communities with exceptionally high modularity ($Q = 0.8049$), indicating strong partitioning into specialized subgroups. Most communities are reviewer-heavy (72-79% reviewers), suggesting taste-based clustering where users with similar preferences review overlapping movie sets. Among highly active users (50+ reviews), rating behavior stratified into critics (14.9%), neutral reviewers (40.4%), and enthusiasts (44.7%), revealing systematic heterogeneity that recommendation algorithms should account for rather than treat all ratings equally.

The taste-change analysis provides a different perspective of the network. Of 5,401 multi-year reviewers, 56.2% exhibited significant rating drift ($|\Delta_i| > 0.5$ stars), with 31.6% becoming more critical and 24.6% less critical over time. This finding suggests opportunities for time-weighted recommendation strategies that prioritize recent reviewer behavior rather than their overall reviewing behavior.

These results directly inform platform design. The degree distributions (heavy-tailed reviewer and movie connectivity) suggest that hub nodes (prolific reviewers and blockbuster movies) disproportionately influence network structure and could be ommitted for analyzing smaller films. The high modularity indicates that taste-based filtering could improve recommendations by exploiting natural community boundaries. The observed taste drift suggests that user models should incorporate temporal dynamics rather than assuming fixed preferences.

4.1 Limitations

I used a 5% uniform random sample for computational tractability, which may underrepresent rare edge patterns. The bipartite structure limits certain analyses (e.g., reviewer-reviewer similarity networks). Future work should check:

- Full-dataset analysis to validate the sampling effects
- Temporal community tracking to identify when and why communities split or merge
- Genre-labeled analysis to test whether communities correspond to known movie categories

4.2 Future Work

There are some future questions that can also be investigated:

1. Can early movie review patterns (i.e. the first 10 or 20 reviews) predict future movie-taste trajectories?
2. Do movie Enthusiasts (high-raters) migrate to different communities than movie Critics (lower-raters)?
3. How has random sampling affected the modularity?
4. How do hub reviewers (e.g. top 1% of reviewers) influence the overall movie taste of their local neighborhoods?

5 Conclusion

In summary, the Amazon movie review network exhibits temporal growth with declining density, a dominant giant component amid fragmentation, strong community partitioning ($Q = 0.8049$), heterogeneous user types, and a measurable taste evolution affecting 56.2% of repeat reviewers. These findings demonstrate that collaborative rating systems are not static databases but evolving social-information networks useful for user movie-preference modeling. The combination of structural metrics, community detection, and behavioral tracking provides a characterization of how large-scale review platforms organize themselves over time.

References

[1] Julian McAuley and Jure Leskovec. "From Amateurs to Connoisseurs: Modeling the Evolution of User Expertise Through Online Reviews". In: *Proceedings of the 22nd International Conference on World Wide Web (WWW)*. 2013. URL: https://snap.stanford.edu/data/web-Movies.html.

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